Results: recovery checks and study suite

Every quantitative assertion in this and the following results sections is a generated token, hydrated by the manuscript-variable generator from analysis outputs produced by `src/analysis/workflow.py` and `src/fedference/experiments/` — no number is transcribed by hand. The studies implement categorical source-mechanism analogues of the colony belief-sharing scenario (Friston et al. 2024) and add the contaminated-sentinel robustness sweep and the federated neural-network baseline that are this paper's robust-federated-learning contribution. All runs are deterministic under seed 0.

Results: recovery checks and study suite I

We lead with the recovery limits, not with a study, because they are what makes the studies a single coherent system rather than a collection of unrelated experiments. The generalized-Bayes machinery of sec. 5 contains the standard-Bayes client corner, while the server has the exact project-local zero-robustness log-linear-pool identity. Under the explicit bridge in sec. 5.8, that pool is a categorical specialization of Eq. 7's message-combination term rather than the complete source protocol. We verify those limited identities to machine precision before reporting anything built on top of them.

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision I

The identities that anchor every result are the client recovery of standard Bayes at the KL/NLL/$\beta \to 0$ and $q_{\text{loss}} \to 0$ limits plus the project-local server recovery to the log-linear pool at $c = 0$ — the scoped claims of sec. 6 (Corollary 6, Lemma 3, Theorem 5). These are not figures but exact equalities, pinned by the locked core test suite. Under the theorem's shared-support, posterior-log-potential, and fixed-weight assumptions, the server pool specializes Eq. 7's message-combination term; it does not reproduce the source construction in full. Robustness is a tested extension that vanishes at the stated recovery limits.

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision I

The five residuals below are the maximum absolute deviations between each generalized-Bayes object and the standard object it must reproduce in the trusting limit. Each is a deterministic constant of the mathematics, not a per-run sample: it is reported as the maximum absolute deviation over the recovery band, which is exactly 0 where the implementation evaluates the closed form at the limit (the Rényi/loss switch) and otherwise a tiny floating-point residual:

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision I

- The server-side aggregator at zero robustness equals the log-linear pool (eq. 7, Theorem 5): maximum absolute deviation 0. This is the *naive-recovery* limit of the server-side heuristic — the only property proven for that axis (see sec. 3.3).
- The generalized posterior under the KL divergence and the NLL loss equals the closed-form prior×likelihood Bayes posterior (eq. 6, Corollary 6): maximum absolute deviation 5.55e-17.
- The Rényi divergence recovers KL as $\alpha \to 1$ (eq. 3, Lemma 3): residual 0.
- The β-loss recovers the NLL as $\beta \to 0$ (eq. 4, Proposition 4): residual 0; and the robust categorical cross-entropy recovers the NLL as $q_{\text{loss}} \to 0$ (eq. 5, Proposition 4): residual 0.

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision I

Because the Rényi divergence and the two categorical losses switch to their exact closed form inside narrow numerical-stability bands around the limit point (the Rényi switch band for α and the categorical-loss switch band for q_{loss} and β), the three zero residuals above confirm that branch equals the standard object — not, by themselves, that the *general* formula converges there. As a genuine (non-branch) convergence witness, evaluating each general formula strictly *outside* its switch band — $q_{\mathrm{loss}} = \beta = 1.00 \times 10^{-6}$ for the categorical losses and $\alpha = 1.00001$ for the Rényi divergence — gives residuals 1.12e-05 (rcce), 1.24e-05 (β-loss), and 1.66e-05 (Rényi): nonzero (a small multiple of the input offset itself, as the first-order Taylor behavior near the limit predicts) yet still several orders of magnitude below the $O(1)$ scale of the loss/divergence values being compared, and shrinking monotonically as the offset shrinks toward the switch band (verified in `tests/fedference/test_core_identities.py`). This is evidence that the general formula itself converges to the standard-Bayes limit, not merely that the implementation switches to it exactly at the corner.

The first residual is the naive-aggregate limit of the *server-side* heuristic (Theorem 5); the latter four are the per-agent generalized-Bayes recoveries (Corollary 6 + Proposition 4) and the divergence-family recovery in the Rényi limit (Lemma 3, eq. 3) that define the theorem-bearing FedGVI axis under matching assumptions. Keeping the three axes distinct at the level of the recovery limits is what lets the robustness claims of sec. 7.5 and sec. 7.6 rest on the per-agent axis without leaning on the heuristic.

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision I

257 of 259 acceptance criteria are verified. The pure-NumPy/SciPy core carries project test coverage of 90.26% (gate $\geq 90\%$), with every stochastic step threaded through a single seeded `np.random.default_rng(0)`. sec. 10 records the full environment fingerprint, and the expected-free-energy identity that underwrites the active-inference substrate is proven and visualized in sec. 6.2 (fig. 7).